Consider the $(nM-n)$ sequences in C' that are cyclically distinct from $a_i(k)$. These may be partitioned into $(M-1)$ equivalence classes by defining two sequences to be equivalent if one is a cyclic shift of the other. The index set B_i is obtained by picking the index of precisely one sequence from each equivalence class.

Using the fact that

$$\sum_{\tau=1}^{n-1}\left(\sum_{k=0}^{n-1}a_i(k)a_i(k\oplus\tau)\right)=\omega(\omega-1)$$

the first summation in the right-hand side of (A21) can be bounded with the previous equation as

$$\sum_{i=1}^{nM}\sum_{\tau=1}^{n-1}\left(\sum_{k=0}^{n-1}a_i(k)a_i(k\oplus\tau)\right)^l \le nM\cdot\left(\frac{\omega(\omega-1)}{\lambda}\right)\cdot\lambda^l. \quad \text{(A22)}$$

Similarly, the second summation in the right-hand side of (A21) has the upper bound

$$\sum_{i=1}^{nM}\sum_{j\in B_i}\sum_{\tau=0}^{n-1}\left(\sum_{k=0}^{n-1}a_i(k)a_j(k\oplus\tau)\right)^l \le nM(M-1)\cdot\left(\frac{\omega^2}{\lambda}\right)\cdot\lambda^l \quad \text{(A23)}$$

since

$$\sum_{\tau=0}^{n-1}\left(\sum_{k=0}^{n-1}a_i(k)a_j(k\oplus\tau)\right)=\omega^2.$$

We note that equality holds in (A22) and (A23) iff the correlation values are distributed such that as many of the correlation values as possible equal λ with the rest being equal to zero that in turn can only happen if $\lambda|\omega(\omega-1)$ and $\lambda|\omega^2$, respectively.

Plugging (A22) and (A23) into (A21) gives

$$(nM-1)m_l \le (M\omega^2-\omega)\lambda^{l-1}. \quad \text{(A24)}$$

Comparing the lower bounds in (A19) and (A20) to the upper bound in (A24) gives the bound in Theorem 2 on the maximum possible value of the family size M. □

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A New Construction of Two-Dimensional Arrays with the Window Property

J. DÉNES AND A. D. KEEDWELL

Abstract—**Tuscan squares, row-complete latin squares and comma-free codes are used to construct binary and nonbinary arrays with a certain window property. Such arrays have practical application in the coding and transmission of pictures.**

I. Introduction and Definitions

A two-dimensional array of zeros and ones is said to have the $u\times v$ $(u,v\ge 2)$ *horizontal window property* if every nonnull pattern of zeros and ones is seen at most once when a window of u rows and v columns is moved horizontally across the array. (To avoid trouble at the ends, the array should be imagined as being written on a vertical cylinder.) Similarly the $u\times v$ *vertical window property* holds if every nonzero pattern appears at most once when the window is moved vertically down the array (and the array is regarded as being written on a horizontal cylinder).

Two-dimensional arrays of zeros and ones that have the stronger property that every nonnull pattern of zeros and ones is seen *exactly* once when a window of u rows and v columns is moved horizontally and vertically across the array have been called *perfect maps* or *pseudo-random arrays* of a special type. Some early papers on the construction of such arrays are [10], [16], [18], [21], and [22]. Most of the constructions used in these and later papers make use of sequences that have a one-dimensional (1-D) window property (de Bruijn sequences).

In this correspondence, we first show how row-complete latin squares and Tuscan squares can be used to construct nonbinary 1-D and 2-D arrays that have the $u\times v$ horizontal and/or vertical window property for all sufficiently large windows. We then show how binary arrays with a similar property can be derived. Since an array can only be a perfect map for one particular size of window, the arrays that we construct are not perfect but nevertheless they have several practical applications, among others in connection with the coding and transmission of pictures, as has been pointed out in [17]. Our method is deterministic rather than stochastic and the arrays that we derive contain an effectively random selection of the windows of each relevant size. We shall require some preliminary definitions.

Definition 1: A nonbinary $m\times n$ array is said to have the $u\times v$ *window property* if all its subarrays of size $u\times v$ are different.

Definition 2: A nonbinary sequence (one dimensional array) is said to have the *window property for windows of width u* if every subsequence of length u is different.

Definition 3: An $n\times n$ matrix defined on the symbols $0,1,\cdots,n-1$ in which each of the symbols occurs once in each row and in which each of the $n(n-1)$ ordered pairs of symbols occurs in adjacent positions in exactly one row is called a *Tuscan square* (S. W. Golomb and H. Taylor [9]). If, in addition, each of the symbols occurs exactly once in each column, the array is

Manuscript received October 20, 1987; revised November 2, 1989. This work was presented at the One-Day Combinatorics Colloquium, University of Reading, Reading, England, April 26, 1989. This work was supported in part by SERC Grant No. GRD 79029.

J. Dénes is at Csaba U10, Budapest 1122, Hungary.

A. D. Keedwell is with the Department of Mathematics, University of Surrey, Guildford, England GU2 5XH.

IEEE Log Number 9034943.

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called a *row-complete latin square* or (by Golomb and Taylor) a *Roman Square*.

Definition 4: A set C of codewords each of length m and defined over an alphabet of q symbols is said to form a *comma-free code*, which we shall denote by CF(q,m), if, for each pair of words $\boldsymbol{a}=(a_1a_2\cdots a_m)$ and $\boldsymbol{b}=(b_1b_2\cdots b_m)$ in C, none of the words $(a_{i+1}a_{i+2}\cdots a_mb_1b_2\cdots b_i)$, where $i=1,2,\cdots,m-1$, is in C.

It will be convenient later in the correspondence to denote the set of all words of the form $(a_{i+1}a_{i+2}\cdots a_mb_1b_2\cdots b_i)$, where i is fixed, by $L_i(C)$ and to call it the ith *overlap* of the code C.

Example 1: The three words (1 0 0 0), (1 0 0 1), (1 0 1 1) form a CF(2,4). Its first overlap is {(0 0 0 1), (0 0 1 1), (0 1 1 1)}. Its second and third overlaps are {(0 0 1 0), (0 1 1 0), (1 1 1 0)}, and {(0 1 0 0), (0 1 0 1), (1 1 0 0), (1 1 0 1)}, respectively.

Example 2: The two words (1 0 0), (1 0 1) form a CF(2,3) and the six words (1 0 1 0 0), (1 0 1 0 1), (1 0 0 0 0), (1 0 0 0 1), (1 0 0 1 1), (1 0 1 1 1) form a CF(2,5).

II. One-Dimensional Arrays

In a Tuscan square, each symbol occurs exactly once in each row and is followed exactly once by each of the other symbols. Consequently, there is exactly one row in which a particular symbol has no successor. This means that each symbol occurs exactly once in the last column of the square. Similarly, each symbol is preceded exactly once by each of the other symbols and so each symbol occurs exactly once in the first column of the square.

Let us take an $n\times n$ Tuscan square T defined on the symbols $0,1,2,\cdots,n-1$ and adjoin to it two additional columns, the first being a duplicate of the last column of T and the second a column all of whose entries are the integer n, and then construct a sequence that consists of the successive rows of the augmented square written one after the other, the rth following the $(r-1)$th and the first following the last after the adjunction of an additional copy of the symbol n. Then it is easy to see that the sequence so obtained is a nonbinary sequence that has the window property for a window of length 2 and consequently also for windows of all lengths greater than 2. An example for the case $n=4$ is shown in the following:

$$T=\left|\begin{array}{cccc} 0&3&1&2\\ 1&0&2&3\\ 2&1&3&0\\ 3&2&0&1 \end{array}\right| \quad \left|\begin{array}{cccccc} 0&3&1&2&2&4\\ 1&0&2&3&3&4\\ 2&1&3&0&0&4\\ 3&2&0&1&1&4&4 \end{array}\right|.$$

The sequence is 0 3 1 2 2 4 1 0 2 3 3 4 2 1 3 0 0 4 3 2 0 1 1 4 4 $\cdots\cdots$.

We observe that such a sequence is perfect relative to a window of length 2 since every ordered pair (u,v) occurs exactly once in the sequence.

If now the $n+1$ different integers which occur in the augmented square are replaced by $n+1$ different codewords of a binary comma-free code CF$(2,m)$, then the resulting binary sequence S consists of $[(n+2)n+1]m=(n+1)^2m$ symbols and has the window property for all windows of width $3m-1$ or greater. To see this, we note that a window of length $w\geq 3m-1$ spans at least one pair (c_u,c_v) of consecutive complete codewords of C that have replaced a corresponding pair (u,v) of consecutive integers in the nonbinary sequence. Since the code CF$(2,m)$ is comma-free, no codeword of CF$(2,m)$ occurs as the first m binary digits of any subsequence of S except one starting at the $(im+1)$th digit of S for some i, $i=0,1,\cdots,(n+2)n$. Consequently, the subsequence $t=c_uc_v$ of binary digits occurs only once in S and is in a different position in each window of length w which spans it. (Note that the codewords of the comma-free code can be assigned to the integers in an arbitrary order and that the Tuscan square to be used can be arbitrarily chosen.)

As examples, we exhibit in the following diagram the complete binary sequence that is constructed from the (unique) row-complete latin square L of order 2 using the three codewords of the comma-free code CF(2,4) given in Example 1. This sequence contains $(2+1)^2\times 4=36$ binary digits and has the window property for all windows of length 11 or greater. (If the window width is 11, there are 26 different windows.) Examples exhibited are

$$L=\left|\begin{array}{cc} 0&1\\ 1&0 \end{array}\right| \quad \left|\begin{array}{ccccc} 0&1&1&2&\\ 1&0&0&2&2 \end{array}\right| \qquad \begin{array}{l} c_0=(1\ 0\ 0\ 0)\\ c_1=(1\ 0\ 0\ 1).\\ c_2=(1\ 0\ 1\ 1) \end{array}$$

The binary sequence is 1 0 0 0 1 0 0 1 1 0 0 1 1 0 1 1 1 0 0 1 1 0 0 0 1 0 0 0 1 0 1 1 1 0 1 1 $\cdots\cdots$.

Next, we exhibit the first part of the binary sequence that is constructed from the Tuscan square T_7 of order 7, which is shown next. (The Tuscan square has been taken from a list given in [9].) The nonbinary digits are replaced by the first eight codewords of the comma-free code CF(2,6) given later in this correspondence. This binary sequence has the window property for all windows of length 17 or greater. For the window width 17, there are $[(7+1)^2\times 6]-16=368$ different windows in the constructed sequence. The binary sequence shown as

$$T=\left|\begin{array}{ccccccc} 0&1&2&3&4&5&6\\ 1&6&4&2&5&3&0\\ 2&4&0&6&5&1&3\\ 3&2&0&5&4&6&1\\ 4&1&5&0&3&6&2\\ 5&2&6&3&1&0&4\\ 6&0&2&1&4&3&5 \end{array}\right| \quad \left|\begin{array}{ccccccccc} 0&1&2&3&4&5&6&6&7\\ 1&6&4&2&5&3&0&0&7\\ 2&4&0&6&5&1&3&3&7 \end{array}\right|$$

$$\begin{array}{l} c_0=(0\ 0\ 0\ 1\ 0\ 0)\\ c_1=(0\ 0\ 1\ 1\ 0\ 0)\\ c_2=(0\ 0\ 1\ 1\ 0\ 1)\\ c_3=(0\ 1\ 0\ 1\ 0\ 0)\\ c_4=(0\ 1\ 1\ 1\ 0\ 0)\\ c_5=(0\ 1\ 1\ 1\ 0\ 1)\\ c_6=(0\ 1\ 0\ 1\ 1\ 0)\\ c_7=(0\ 1\ 1\ 1\ 1\ 0). \end{array}$$

The first part of the sequence is 0 0 0 1 0 0 0 0 1 1 0 0 0 0 1 1 0 1 0 1 0 1 0 0 0 1 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 0 0 1 0 1 1 0 0 1 1 1 1 0 0 0 1 1 0 0 0 1 0 1 1 0 0 1 1 1 0 0 0 0 1 1 0 1 0 1 1 1 0 1 $\cdots\cdots$.

III. Two-Dimensional Arrays

We turn next to two-dimensional (2-D) arrays with the window property. We observe firstly that any Tuscan square or row complete latin square of order n $(n>2)$ or any $r\times n$ subrectangle of such a square ($r<n$ and $n>2$) is a nonbinary array with the window property for all $u\times v$ windows with $v\geq 2$. The same statement remains true if two additional columns are adjoined

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to the Tuscan square as our construction of 1-D arrays.

It follows immediately that, if the h $(n \le h \le n+1)$ different integers that occur in the square or subrectangle (or augmented square with $n+2$ columns) are replaced by h different codewords of a binary comma-free code CF$(2,m)$, then we shall obtain a zero-one-array of size $r \times (n+2)m$ (in the case of the augmented square) which has the $u \times v$ window property for all $v \ge 3m-1$. (If and only if $v \ge 3m-1$, each $1 \times v$ window will span completely the binary representation of some two consecutive nonbinary digits and, since the binary representation for every two consecutive digits is different, the result follows as for sequences.)

As an illustration, we first exhibit a row-complete latin square of order 6 to which an additional column has been adjoined. We then exhibit the 3×35 rectangle of zeros and ones derived from its first three rows using the comma-free code CF$(2,5)$ of Example 2. The latter rectangle has the $u \times v$ window property for all $v \ge 14$ and for all values of u. First, we exhibit

$$\begin{array}{|ccccccc|}\hline 0&1&5&2&4&3&3\\ 1&2&0&3&5&4&4\\ 2&3&1&4&0&5&5\\ 3&4&2&5&1&0&0\\ 4&5&3&0&2&1&1\\ 5&0&4&1&3&2&2.\\ \hline\end{array}$$

Secondly, we exhibit

$$\begin{array}{|l|}\hline 10100101011011110000100111000110001\\ 10101100001010010001101111001110011\\ 10000100011010110011101001011110111.\\ \hline\end{array}$$

IV. Existence Questions

For the implementation of the previous constructions, it is necessary to know for which orders n Tuscan squares exist and also to be able to construct binary comma-free codes CF$(2,m)$ and to know how many codewords each contains.

It has been shown that Tuscan squares exist of each finite order n except 3 and 5. Nonexistence for the orders 3 and 5 is easy to check, see, for example, [9].

For even orders n, a very simple construction of a row-complete latin square has been given by E. J. Williams [25] and, independently, by several other authors. (See Ch. 2 of [1].) The first row of the square is $0\ \ 2m-1\ \ 1\ \ 2m-2\ \ 2\ \ \cdots\ \ \cdots\ \ 2m-1-(m-1)\ \ m-1\ \ m$ and the $(r+1)$th row is obtained by adding r modulo $2m$ to the elements of the first row, $0 \le r \le 2m-1$. (An alternative first row is $0\ \ 1\ \ 2m-1\ \ 2\ \ 2m-2\ \ \cdots\ \ \cdots\ \ m-1\ \ 2m-(m-1)\ \ m$ as in the row-complete latin square of order 6.)

For odd orders n, a solution has been given by T. W. Tillson [24]. Tillson has shown that for $2m \ge 8$, the edges of the complete directed graph K^*_{2m-1} on $2m-1$ vertices can be partitioned into $2m-1$ directed Hamiltonian paths. If we label the vertices of K^*_{2m-1} by the symbols $0,1,\cdots,2m-2$, these Hamiltonian paths define the rows of a Tuscan square.

It is known [8] that the number of words in a comma-free code CF(q,m) of word length m cannot exceed

$$s(q,m) = \frac{1}{m}\sum_{d|m} q^{m/d}\mu(d),$$

where $\mu(d)$ is the Mobius function defined as

$$\mu(d) = \begin{cases} 1, & \text{if } d = 1,\\ 0, & \text{if } a^2 | d \text{ for some } a > 1,\\ (-1)^t, & \text{if } d \text{ is a product of } t \text{ distinct primes.}\end{cases}$$

Moreover, when m is any odd integer, a constructive method for obtaining a comma-free code with exactly $s(q,m)$ codewords has been given by W. L. Eastman [2]. For the case when m is even, rather less is known. However, there exist binary comma-free codes CF$(2,2h)$ of maximal size $s(2,2h)$ in the cases $h = 2,3,4,5$. A maximal CF$(2,4)$ is given in Example 1. A maximal CF$(2,6)$ was first obtained by B. H. Jiggs [12] and, according to Jiggs, the first person to exhibit a maximal CF$(2,8)$ was R. Jewett (unpublished). A maximal CF$(2,10)$ was obtained by Y. Niko [20]. It is known that, for some values of q and m, maximal CF(q,m)'s do not exist when m is even. In particular, it has been proved in [12] that, if $m = 2h$, then a maximal CF(q,m) does not exist if $q > 2^h + h$. A recent improvement on this result is in [23].

The values of $s(2,m)$ for $m = 2,3,4,5,6,7,8,9,10$ are $1,2,3,6,9,18,30,56,99$. More generally, $s(2,m) \approx 2^m/m$ for large m.

When m is even, the following result, which is stated in [12] without proof, gives a lower bound for the maximal number of words in a CF(q,m).

Theorem: Let $m = 2h$ and let D be a comma-free code CF$(q,2)$ in which the codewords have length 2 and that contains q^2/t codewords. Let X denote the set of all words of length 2 constructed from the alphabet of q symbols, so that $|X| = q^2$. Suppose that $D = \{(\alpha_1\beta_1),(\alpha_2\beta_2),\cdots\}$ and that $X - D = \{(a_1b_1),(a_2b_2),\cdots\}$. Then the set of words

$$\boldsymbol{c}_j = (\alpha_j\ a_{j_2}\ a_{j_3}\ \cdots\ a_{j_h}\ \beta_j\ b_{j_2}\ b_{j_3}\ \cdots\ b_{j_h})$$

form a comma-free code CF$(q,2h)$ that contains $(q^m/2)(1-1/t)^{h-1}$ codewords. The number of codewords is largest when $t = h$, and the number of codewords is then $\lfloor q^2/h \rfloor \cdot \lceil q^2 - q^2/h \rceil^{h-1}$.

Proof: First we note that $X - D$ contains $q^2 - q^2/t$ words. It follows that there are $(q^2 - q^2/t)^{h-1}$ ways of choosing $h-1$ words from $X - D$ to use in the construction of a codeword $\boldsymbol{c}_j$. This gives a total of $(q^2/t)(q^2 - q^2/t)^{h-1} = (q^m/t)(1-1/t)^{h-1}$ codewords for our comma-free code. It is easy to see that $(1/t)(1-1/t)^{h-1}$ is maximal when $t = h$. Since h is an integer that may not divide q^2, the number of codewords is then $\lfloor q^2/h \rfloor \cdot \lceil q^2 - q^2/h \rceil^{h-1}$.

Since a maximal CF$(q,2)$ contains $\lfloor q^2/3 \rfloor$ codewords (see [8]), the construction can always be carried out with $t = h$ provided that $h \ge 3$.

Next, we wish to show that the code C that we have constructed is comma-free. Let us consider the words that appear in the ith overlap $L_i(C)$, $i < h$. $L_i(C)$ contains words of the form

$$a_{j_{i+1}}\ a_{j_{i+2}}\ \cdots\ a_{j_h}\ \beta_j\ b_{j_2}\ b_{j_3}\ \cdots\ b_{j_h}\ \alpha_j\ a_{j_2}\ a_{j_3}\ \cdots\ a_{j_i}.$$

Since $(a_{j_{i+1}}\ b_{j_{i+1}}) \notin D$, this is not a codeword of C. Similarly, the words that appear in the ith overlap for $i > h$ are not in C. The words that appear in the hth overlap $L_h(C)$ take the form

$$\beta_j\ b_{j_2}\ b_{j_3}\ \cdots\ b_{j_h}\ \alpha_j\ a_{j_2}\ a_{j_3}\ \cdots\ a_{j_h}.$$

Since D is comma-free, $(\beta_j\ \alpha_j)$ is not in D, so words of $L_h(C)$ are not in C.

Hence, for all $h \ge 3$, there exists a CF$(q,2h)$ that contains at least $\lfloor q^2/h \rfloor \cdot \lceil q^2 - q^2 \rceil^{h-1}$ codewords.

When $q = 2$ and $h = 3$, $\lfloor q^2/h \rfloor \cdot \lceil q^2 - q^2/h \rceil = 9$ and so we can construct a maximal CF$(2,6)$. This was the method used by

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Jiggs [12]. We may take $D = \{(0\ 1)\}$ and $X - D = \{(0\ 0), (1\ 0), (1\ 1)\}$. There are nine ways of selecting two, not necessary distinct, pairs from $X - D$ (where the order of selection is significant) and we thus get the maximal CF(2,6) with codewords (0 0 0 1 0 0), (0 0 1 1 0 0), (0 0 1 1 0 1), (0 1 0 1 0 0), (0 1 1 1 0 0), (0 1 1 1 0 1), (0 1 0 1 1 0), (0 1 1 1 1 0), (0 1 1 1 1 1), which we used previously.

We end this account by emphasizing again that a number of algorithmic constructions of maximal length de Bruijn sequences and cycles are known. In particular, one of our referees has cited [7], [14], [15], and [19]. It is not the purpose of this correspondence to parallel these results but rather to suggest a means of constructing a series of pseudo-random arrays (with good randomness characteristics) suitable for picture coding.

We also draw the reader's attention to three recent papers on the subject of the construction of perfect maps not mentioned in the above discussion; namely, [3], [5], and [13]; and to a very recent preprint [4] in which T. Etzion has made use of constructions from [9], [24], and [25] to obtain many new Tuscan squares of both odd and even orders.

We learned recently that A. Iványi has also contributed papers on two-dimensional arrays with the window property in which he has looked at the subject from a different point of view (see, for example [11]). Also, at a recent conference held after this correspondence was prepared for publication, G. Fazekas gave a lecture in which he exploited an idea, also proposed independently by S. L. Ma [17] and in this correspondence, of encrypting a picture by superimposing on it a nonbinary array with the window property. (See Definition 1.) Further details are to be published in a forthcoming paper [6].

Finally, we remark that the reader seeking further information about row-complete and other types of latin squares will find it in [1] and in a forthcoming new book by the same authors.

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Algebraic Decoding of the (32, 16, 8) Quadratic Residue Code

IRVING S. REED, FELLOW, IEEE, XIAOWEI YIN, AND TREIU-KIEN TRUONG, SENIOR MEMBER, IEEE

***Abstract*—An algebraic decoding algorithm for the 1/2-rate (32, 16, 8) quadratic residue (QR) code is found. The key idea of this algorithm is to find the error locator polynomial by a systematic use of the Newton identities associated with the code syndromes. The techniques developed here extend the algebraic decoding algorithm found recently for the (32, 16, 8) QR code. It is expected that the algebraic approach developed here and by Elia apply also to longer QR codes and other BCH-type codes that are not fully decoded by the standard BCH decoding algorithm.**

I. Introduction

It is well known that the quadratic residue codes are good codes of large distance with a rate close to 1/2. If an overall parity check is used, the rate is exactly 1/2. Apparently the algebraic decoding algorithms for the QR codes have not been considered previously with much seriousness. The following nonalgebraic techniques have been considered for these and other cyclic codes [2]: the permutation method, Kasami's error trapping method, the covering polynomial technique and the t-design decoding method. However, the previous techniques have limitations due either to excessively large permutation sets or to the difficulty of finding the proper covering polynomials.

A new decoding technique for the (23, 12, 7) Golay code is given by Elia [1]. More recently P. Bours *et al.* extended this technique to certain other cyclic codes [5] but not to the QR codes. These latter techniques are similar but do not go as far as the algebraic approach described here for finding the coefficients of an error-locator polynomial by directly solving the Newton identities. In this correspondence the algebraic methods similar to those of Elia are used to correct the three possible errors in the (31, 16, 7) QR code. Finally this decoding procedure is extended to the 1/2 rate (32, 16, 8) QR code. This new decoder is based on the fact that the required error-locator

Manuscript received October 1989; revised November 22, 1989.

I. S. Reed and X. Yin are with the Department of Electrical Engineering, University of Southern California, Los Angeles, CA 90089-0271.

T. K. Truong is with the Communication System Research Section, Jet Propulsion Laboratory, Pasadena, CA 91109.

IEE Log Number 9034946.

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